Activity – 8 : Applying the Route Finding Problem to find the path cost of a given graph

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1. Aim

To apply the route finding problem to the road graph given in the session, identify every possible route from the start city **Coimbatore** to the goal city **Chennai**, calculate the path cost of each route, and determine the optimal (least-cost) path.

2. Problem Formulation

A route finding problem is a search problem defined by five components:

- **Initial state:** Coimbatore
- **Goal state:** Chennai
- **Actions:** Travel from the current city to a directly connected next city (following the direction of the arrows).
- **Transition model:** Moving along an edge changes the current state to the city at the end of that edge.
- **Path cost:** The sum of the edge costs (distances) along the route. The step cost of each edge is the number shown on it in the graph.

A **solution** is any sequence of actions leading from Coimbatore to Chennai. An **optimal solution** is the one with the lowest total path cost.

3. The Graph (Edge Costs)

The graph is directed. The edges and their costs are:

From	To	Cost
Coimbatore	Mettupalayam	37
Coimbatore	Palladam	38
Coimbatore	Pollachi	37

From	To	Cost
Coimbatore	Erode	100
Palladam	Kangayam	32
Kangayam	Karur	61
Karur	Trichy	74
Pollachi	Palani	60
Palani	Dindigul	57
Dindigul	Trichy	97
Trichy	Chennai	320
Erode	Salem	66
Salem	Vellore	200
Vellore	Chennai	140

Mettupalayam has no outgoing edge, so it is a dead end and cannot lead to the goal.

4. Possible Routes and Path Costs

Route 1 (upper route, via Palladam): Coimbatore → Palladam → Kangayam → Karur → Trichy → Chennai Path cost = 38 + 32 + 61 + 74 + 320 = **525**

Route 2 (middle route, via Pollachi): Coimbatore → Pollachi → Palani → Dindigul → Trichy → Chennai Path cost = 37 + 60 + 57 + 97 + 320 = **571**

Route 3 (lower route, via Erode): Coimbatore → Erode → Salem → Vellore → Chennai Path cost = 100 + 66 + 200 + 140 = **506**

Route 4 (via Mettupalayam): Coimbatore → Mettupalayam: dead end, no path to Chennai.

Route	Path	Number of edges	Total cost

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1	Coimbatore – Palladam – Kangayam – Karur – Trichy – Chennai	5	525
2	Coimbatore – Pollachi – Palani – Dindigul – Trichy – Chennai	5	571
3	Coimbatore – Erode – Salem – Vellore – Chennai	4	506
4	Coimbatore – Mettupalayam	1	No path to goal

5. Uniform Cost Search Trace

Uniform Cost Search (UCS) always expands the frontier node with the lowest path cost g(n), so it is guaranteed to return the optimal path.

Step	Node expanded	g(n)	Nodes added to frontier (cost)
1	Coimbatore	0	Mettupalayam (37), Palladam (38), Pollachi (37), Erode (100)
2	Mettupalayam	37	none (dead end)
3	Pollachi	37	Palani (97)
4	Palladam	38	Kangayam (70)
5	Kangayam	70	Karur (131)
6	Palani	97	Dindigul (154)
7	Erode	100	Salem (166)
8	Karur	131	Trichy (205)
9	Dindigul	154	Trichy (251), discarded because 205 is cheaper

Step	Node expanded	g(n)	Nodes added to frontier (cost)
10	Salem	166	Vellore (366)
11	Trichy	205	Chennai (525)
12	Vellore	366	Chennai (506), better than 525, so it replaces it
13	Chennai	506	Goal reached

6. Result

The optimal path found is:

Coimbatore → Erode → Salem → Vellore → Chennai

Minimum path cost = 100 + 66 + 200 + 140 = 506

7. Observations

- The lowest-cost route (506) is not the one with the cheapest first step. Coimbatore → Erode costs 100, the most expensive first edge, yet the route is cheapest overall. This shows why a greedy choice of the nearest next city does not always give the best path, and why path cost must be accumulated over the whole route.
- The Trichy route is expensive mainly because of the final Trichy → Chennai edge (320).
- Route 3 has fewer edges (4) than Routes 1 and 2 (5 each), but fewer edges does not always mean lower cost; here it happens to be both.
- Cost difference: Route 1 is 19 more than the optimum (525 − 506), and Route 2 is 65 more (571 − 506).

8. Conclusion

By formulating the graph as a route finding problem and applying Uniform Cost Search, all feasible routes from Coimbatore to Chennai were evaluated. The least-cost path is **Coimbatore → Erode → Salem → Vellore → Chennai with a total path cost of 506**.